

DELIVERY-AWARE SPARSE CAUSAL ROUTING FOR DISTRIBUTED MULTI-OBJECT TRACKING

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ABSTRACT

We study sparse communication for distributed labeled multi-Bernoulli tracking when mobile formations change their physical neighbors. A causal routing policy retains a feasible formation tree, repairs it only when necessary, and combines local directed cycles with bidirectional gateways. The scheduled backbone uses $N + 2(F - 1)$ posterior messages for N sensors in F formations. We separate planned connectivity from delivered, label-specific fusion. Paired 160-step development experiments with 24 and 36 sensors reduce attempted posterior bytes by 10.0% and 21.8% relative to fixed routing. Mean OSPA decreases by 2.40% and 0.37%, respectively; larger conditional localization gains do not imply comparable target-set recovery. A post-hoc error budget shows that fixed-cardinality localization improvements alone can reduce mean OSPA by at most 0.28 and 1.31 m in the sparse arms. A receiver ablation exposes a precision-consistency tradeoff.

Index Terms— Multi-object tracking, labeled multi-Bernoulli, distributed fusion, dynamic topology, packet loss

1. INTRODUCTION

Mobile formations exchange posteriors to track beyond individual fields of view. Motion breaks routes; redundant inputs consume bandwidth. Which messages support both object existence and location within limited rounds?

The labeled multi-Bernoulli (LMB) filter represents object existence and spatial states [1]. Consensus LMB filtering uses Kullback–Leibler averaging (KLA), or normalized geometric fusion [2, 3]. Unlike repeated consensus on fixed inputs, tracking interleaves observations with few communication rounds; scheduled connectivity guarantees neither delivery nor lower error.

Randomized gossip relates averaging time to matrix spectra [4]; event-triggered Bayesian filtering uses posterior divergence from a predicted last transmission [5].

Ramachandran et al. jointly reconfigure topology and fusion weights after sensor-quality degradation, then reposition robots for coverage [6]. We do not claim route repair as new. With prescribed motion, we study sparse messages, directed losses and finite-round LMB fusion under unequal views.

Multi-view methods include GM-PHD compensation [7], LMB fusion [8], and per-label weighting [9]. Label matching [10], LMB minimum information loss [11], separate cardinality/localization objectives [12], and exclusive-track handling [13] are established estimator ideas. All our routing arms share an implementation-specific absent-label FoV censor, not these complete algorithms. We claim no new fusion objective.

We distinguish sparse scheduling, delivered weights and target-set recovery through exact message counts, conditional sensitivity bounds and paired full/sparse comparisons. One opened seed per scale reveals development tradeoffs, not generalization.

2. SPARSE CAUSAL LMB ROUTING

2.1. Fusion and communication model

Sensor i maintains an LMB posterior $\pi_i = \{(r_i^\ell, p_i^\ell)\}_{\ell \in \mathbb{L}_i}$, where r_i^ℓ is the existence probability and $p_i^\ell(x)$ is a normalized spatial density. For a common active label and weights $w_j \geq 0$, $\sum_j w_j = 1$, exact KLA satisfies [3]

$$\eta^\ell = \int \prod_j p_j^\ell(x)^{w_j} dx, \quad \bar{p}^\ell(x) = \frac{\prod_j p_j^\ell(x)^{w_j}}{\eta^\ell}, \quad (1)$$

$$\text{logit } \bar{r}^\ell = \sum_j w_j \text{logit } r_j^\ell + \log \eta^\ell. \quad (2)$$

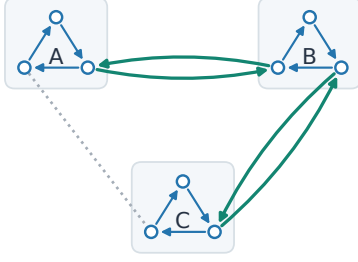
Hölder’s inequality gives $\eta^\ell \leq 1$: spatial disagreement can lower existence log odds relative to their weighted input value. Routing changes both information access and density compatibility.

Let G_t^{phys} describe current physical links, and let A_t be the scheduled directed graph, with rows denoting receivers and columns senders. The controller observes current positions, link reliabilities and its previous route,

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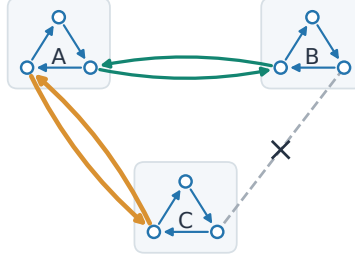
a Retain the feasible tree

$t - 1$: AB + BC



b Repair the failed branch

t : retain AB; add AC



c Delivery is not guaranteed

t : one C-to-A packet lost

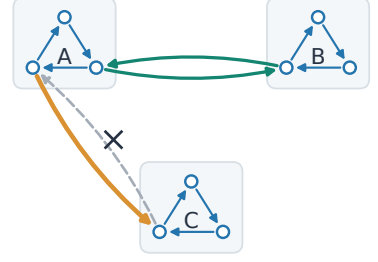


Fig. 1. Sparse causal repair with three illustrative formations of three sensors (not an M24/X36 layout). Blue arrows are local cycles. (a) Retain green AB–BC; dotted AC is unused. (b) Replace failed BC by orange AC, retaining AB. Both plans use 13 messages and at most two nonself inputs per receiver. (c) One C-to-A loss leaves 12 delivered messages and no path out of C. Scheduled connectivity therefore does not ensure delivery, let alone label-specific fusion quality.

but no target truth or future trajectory. The controller uses network-wide current geometry; its decentralized acquisition and dissemination are outside the measured posterior-payload budget. Fusion is local.

2.2. Causal tree repair and sparse inputs

For F formations, a formation edge is usable when current sensor pairs realize its gateways within receiver input constraints. The policy retains the preceding formation tree when feasible. Otherwise it selects a feasible tree lexicographically: minimize replaced formation edges, maximize bottleneck reliability, maximize total log reliability, then minimize distance. Local cycles and gateway endpoints use current geometry; only the formation tree is retained (Fig. 1).

The sparse policy retains one directed cycle inside each formation. Each undirected edge of the formation tree is realized by one directed gateway in each direction; the two directions need not use the same sensor pair. Each receiver has one dominant local input (weight 0.70); gateway receivers also have a cross-formation input (0.05). Remaining weight is assigned to self. The full-repair reference has two nonself inputs per receiver; removing its unused local residual returns 0.05 to self. Proposition 1 (scheduled backbone). If every local cycle and every directed gateway is physically feasible, the scheduled sensor graph is strongly connected and contains exactly

$$M = N + 2(F - 1) \quad (3)$$

directed messages per round. Proof. The cycles contribute N edges and directed paths within each formation. The tree contributes $2(F - 1)$ disjoint gateway edges; concatenating local paths and gateways connects

every ordered sensor pair. This architecture-specific count is neither a global lower bound over strongly connected digraphs nor a guarantee about delivered links.

3. PACKET DELIVERY AND FINITE-ROUND FUSION

3.1. Where a missing weight goes

Let W be a planned row-stochastic matrix with positive diagonal. For receiver i , let $D_{ij} \in \{0, 1\}$ indicate delivery, set $D_{ii} = 1$, and define missing weight mass $m_i = \sum_{j: D_{ij}=0} W_{ij}$. The current reference implementation omits an unavailable input and uses

$$R_{ij} = \frac{W_{ij}D_{ij}}{1 - m_i}. \quad (4)$$

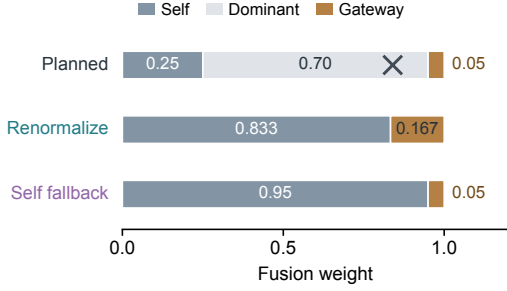
This differs from the planned removal of a local edge. An existing alternative receiver rule returns missing mass to self:

$$S_{ij} = W_{ij}D_{ij} + m_i \mathbf{1}\{i = j\}. \quad (5)$$

Both are row stochastic with identical row perturbation norm $\|R_{i:} - W_{i:}\|_1 = \|S_{i:} - W_{i:}\|_1 = 2m_i$. Figure 2 shows how renormalization amplifies a surviving input, changing overlap and existence. Self fallback avoids amplification but can slow useful information transfer; neither rule is uniformly preferable.

Neither rule preserves double stochasticity under arbitrary directed loss. A symmetric two-node matrix with off-diagonal weight a becomes $\begin{bmatrix} 1 & 0 \\ a & 1-a \end{bmatrix}$ under one-way loss and self fallback; column sums are $1 + a$ and $1 - a$. Reciprocal scheduled gateways therefore do not establish equal-weight KLA consensus.

a A missing packet changes the weights



b Spatial overlap changes existence

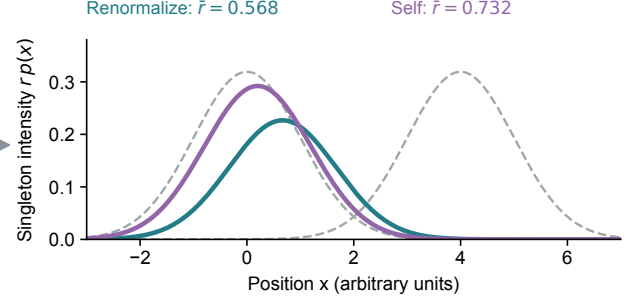


Fig. 2. Packet-to-fusion mechanism. (a) A lost dominant packet changes surviving weights. (b) Exact Bernoulli KLA for surviving Gaussian inputs (dashed), with $r = 0.8$, variance 1 and means 0/4. Weights affect location and existence through overlap. No truth is specified: higher existence does not establish higher accuracy.

3.2. A finite-round sensitivity bound

Freeze positive densities q_j on a common support between fusion rounds, and write $q_\alpha \propto \prod_j q_j^{\alpha_j}$ for a probability vector α . Assume $\text{osc}(\log q_j) = \sup \log q_j - \inf \log q_j \leq L$. Let $P = \widetilde{W}_K \cdots \widetilde{W}_1$ and $Q = W_K \cdots W_1$ be delivered and planned products. For either receiver rule,

$$\begin{aligned} D_{\text{KL}}(q_{P_i} \| q_{Q_i}) &\leq L \|P_i - Q_i\|_1 \\ &\leq 2L \sum_{k=1}^K \max_j m_{j,k}. \end{aligned} \quad (6)$$

Indeed, $h = \sum_j (\alpha_j - \beta_j) \log q_j$ has oscillation at most $L \|\alpha - \beta\|_1$; the log normalizer ratio is in $[\inf h, \sup h]$. Telescoping gives a row bound of $\sum_k \|\widetilde{W}_k - W_k\|_\infty = 2 \sum_k \max_j m_{j,k}$. This elementary bound favors neither receiver rule. Unbounded Gaussian support, mixture truncation, changing labels and intervening observations prevent its direct use as a guarantee for the recursive tracker.

4. EXPERIMENTS AND REMAINING TRADEOFFS

4.1. Paired protocol

M24/X36 have four/six six-sensor formations and 16/24 targets over 160 steps, with overlapping target handovers and route failures. Formation members share a heading, 120° FoV and 300 m range. Nominal detection probability, clutter rate and noise are 0.88, 4 and 7 m; sensing degrades with range and off-axis angle. Seed 1301 pairs truth, measurements, directed delivery uniforms and filter randomness within each scale.

Fixed routing keeps its initial formation tree and re-computes gateways while feasible; otherwise it retains the still-physical initial sensor edges. Full repair retains

two nonself inputs, and sparse repair uses (3); both may repair the tree. All three use (4). The X36 self-fallback ablation changes only handling of unavailable or empty neighbor inputs to (5).

All arms share approximate powered-GM fusion (three components/input, eight output components, 256 tuples); overlap above one triggers moment-matched fallback. Arbitrary GM powers are not exact. Detection probability is evaluated at component means, without boundary splitting [14]. Present labels are not removed solely for being outside the FoV. Each step uses one fusion round, shared birth labels without matching, MAP-cardinality output and existence pruning below 0.01.

Metrics are Euclidean position OSPA (E-OSPA, order 2, cutoff 150 m) [15] and Hungarian-matched position RMSE, averaged over sensor-time cells. RMSE excludes nonempty-empty comparisons and must be read with absolute cardinality error. Focus consistency is pairwise E-OSPA among one fixed representative per formation over $t = 40$ –140, not all-node agreement. Attempted posterior bytes include failed transmissions but exclude routing-control traffic.

4.2. What the comparisons establish

Table 1 and Fig. 3 separate repair from sparsification. Full repair lowers E-OSPA, RMSE and disagreement but costs more bytes. Sparse repair reduces traffic; on X36 its lower RMSE coexists with worse count error and disagreement. Table 2 retains common-cell and formation-tail checks: localization gains survive the former, but means do not establish uniform benefits. Self fallback improves X36 set error and disagreement while worsening common-cell and formation-tail RMSE.

Table 1. Paired 160-step episodes, seed 1301. E-OSPA, conditional RMSE and representative disagreement are in metres; Count is mean absolute cardinality error. All errors and attempted posterior payload (decimal MB) are lower-is-better. Messages are scheduled, before delivery.

Scale	Routing policy	E-OSPA	RMSE	Count	Disagr.	MB	Msgs/step
M24	Fixed formation tree	125.478	22.640	11.131	133.599	40.769	46–48
	Full causal repair	122.380	14.081	10.671	131.913	44.867	48
	Sparse causal repair	122.462	12.183	10.720	131.664	36.676	30
X36	Fixed formation tree	132.680	36.925	18.456	139.407	76.871	66–72
	Full causal repair	131.795	19.586	18.463	139.004	81.259	72
	Sparse causal repair	132.192	19.329	18.597	140.489	60.090	46
	Sparse + self fallback	131.961	20.263	18.518	139.273	60.317	46

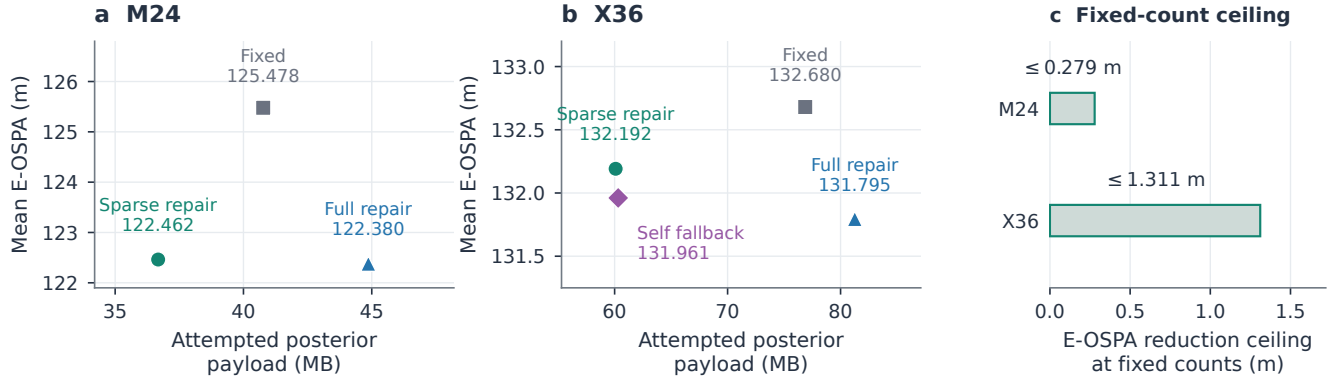


Fig. 3. Communication savings and the remaining target-set gap. (a,b) Payload versus E-OSPA for Table 1; lower left is better, with different vertical scales. Each point is one development episode, without across-seed uncertainty. (c) Upper bounds on further E-OSPA reduction from localization alone in the sparse arms, holding output counts fixed. These bounds are not achieved gains or confidence intervals.

Table 2. RMSE checks (%). Negative gains are worse. Common cells have finite RMSE in both arms, not necessarily identical matches. Formation gains use each arm’s finite cells.

Comparison	Common cells	RMSE gain	
		Common	Worst form.
M24 sparse/fixed	99.870	+46.190	−24.085
X36 sparse/fixed	97.465	+38.431	+4.444
X36 self/sparse	96.493	−5.471	−37.361

4.3. How much can localization alone improve?

To separate position refinement from target-set recovery, write squared OSPA at one sensor–time cell as

$$E^2 = C + L, \quad C = c^2 \frac{|n - T|}{\max(n, T)}, \quad L \geq 0, \quad (7)$$

where T, n are true and estimated counts and $c = 150$ m; empty–empty gives zero. Perfect localization leaves $\sqrt{C}$.

Saved count errors omit sign; OSPA/RMSE constraints give $C_{\min} \leq C \leq C_{\max}$, with 0/3840 M24 and 121/5760 X36 sparse cells unresolved. Count terms account for 99.567% and 98.074–99.337% of summed squared OSPA, not mean OSPA. Figure 3(c) gives the fixed-count ceiling $\text{mean}(E - \sqrt{C_{\min}})$, not an achieved gain. Substantial improvement requires target-set recovery, not just position refinement.

5. CONCLUSION

Sparse repair saves posterior traffic but retains target-set and formation tradeoffs. Useful existence evidence and spatial compatibility remain joint constraints. Multi-view and label-matched baselines, independent seeds and layouts, control traffic and decentralized routing remain open.

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6. REFERENCES

- [1] Stephan Reuter, Ba-Tuong Vo, Ba-Ngu Vo, and Klaus Dietmayer, “The labeled multi-bernoulli filter,” *IEEE Transactions on Signal Processing*, vol. 62, no. 12, pp. 3246–3260, 2014.
- [2] Giorgio Battistelli and Luigi Chisci, “Kullback–leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability,” *Automatica*, vol. 50, no. 3, pp. 707–718, 2014.
- [3] C. Fantacci, B.-N. Vo, B.-T. Vo, G. Battistelli, and L. Chisci, “Consensus labeled random finite set filtering for distributed multi-object tracking,” *arXiv:1501.01579*, 2015.
- [4] S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah, “Randomized gossip algorithms,” *IEEE Transactions on Information Theory*, vol. 52, no. 6, pp. 2508–2530, 2006.
- [5] Giorgio Battistelli, Luigi Chisci, Lin Gao, and Daniela Selvi, “Event-triggered distributed Bayes filter,” *arXiv:1902.09825*, 2019.
- [6] Ragesh Kumar Ramachandran, Nicole Fronda, and Gaurav S. Sukhatme, “Resilience in multirobot multitarget tracking with unknown number of targets through reconfiguration,” *IEEE Transactions on Control of Network Systems*, vol. 8, no. 2, pp. 609–620, 2021.
- [7] Guchong Li, Giorgio Battistelli, Wei Yi, and Lingjiang Kong, “Distributed multi-sensor multi-view fusion based on generalized covariance intersection,” *Signal Processing*, vol. 166, pp. 107246, 2020.
- [8] Suqi Li, Giorgio Battistelli, Luigi Chisci, Wei Yi, Bailu Wang, and Lingjiang Kong, “Multi-sensor multi-object tracking with different fields-of-view using the LMB filter,” in *2018 21st International Conference on Information Fusion (FUSION)*, 2018, pp. 1201–1208.
- [9] Xiaoying Wang, Amirali K. Gostar, Tharindu Rathnayake, Benlian Xu, Alireza Bab-Hadiashar, and Reza Hoseinnezhad, “Centralized multiple-view sensor fusion using labeled multi-bernoulli filters,” *Signal Processing*, vol. 150, pp. 75–84, 2018.
- [10] Suqi Li, Giorgio Battistelli, Luigi Chisci, Wei Yi, Bailu Wang, and Lingjiang Kong, “Computationally efficient multi-agent multi-object tracking with labeled random finite sets,” *IEEE Transactions on Signal Processing*, vol. 67, no. 1, pp. 260–275, 2019.
- [11] Lin Gao, Giorgio Battistelli, and Luigi Chisci, “Fusion of labeled RFS densities with minimum information loss,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 5855–5868, 2020.
- [12] Murat Uney, Jeremie Houssineau, Emmanuel Delande, Simon J. Julier, and Daniel E. Clark, “Fusion of finite-set distributions: Pointwise consistency and global cardinality,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 6, pp. 2759–2773, 2019.
- [13] Lin Gao, Giorgio Battistelli, and Luigi Chisci, “Fusion of labeled RFS densities with different fields of view,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 6, pp. 5908–5924, 2022.
- [14] Keith A. LeGrand and Silvia Ferrari, “Split happens! imprecise and negative information in Gaussian mixture random finite set filtering,” *Journal of Advances in Information Fusion*, vol. 17, no. 2, pp. 78–96, 2022.
- [15] Dominic Schuhmacher, Ba-Tuong Vo, and Ba-Ngu Vo, “A consistent metric for performance evaluation of multi-object filters,” *IEEE Transactions on Signal Processing*, vol. 56, no. 8, pp. 3447–3457, 2008.